

# Le (Leonard) Zhang

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## Education

- Ph.D. in Computer Science**, University of Michigan, Ann Arbor Expected May 2029  
Advisor: Ke Sun
- M.S. in Computer Science**, University of California, San Diego June 2025  
Advisor: Tajana Šimunić Rosing
- B.A. in Computer Science**, University of North Carolina at Chapel Hill December 2022  
Major in Computer Science and Minor in Mathematics  
Advisor: Shahriar Nirjon

## Research Interests

- Intelligent embedded systems, cyber-physical systems, networked systems, and mobile computing.
- Resource-efficient AR/VR/wearable vision systems, AI glasses: networking, perception, on-device learning.
- Multimodal LLM agents, lifelong agentic memory, and efficient VLA systems.

## Conference Publications

1. **Le Zhang\***, Quanling Zhao\*, Run Wang, Shirley Bian, Onat Gungor, Flavio Ponzina, and Tajana Rosing. "Offload Rethinking by Cloud Assistance for Efficient Environmental Sound Recognition on LPWANs." *The ACM Conference on Embedded Networked Sensor Systems (SenSys)*, 2025. (\* for Equal Contribution, Full Paper, Core Rank A\*, Acceptance Rate: 19.2%)
2. **Le Zhang**, Onat Gungor, Flavio Ponzina, and Tajana Rosing. "E-QUARTIC: Energy Efficient Edge Ensemble of Convolutional Neural Networks for Resource-Optimized Learning." *The Asia and South Pacific Design Automation Conference (ASP-DAC)*, 2025. (Core Rank B, Acceptance Rate: 31%)
3. Yubo Luo, **Le Zhang**, Zhenyu Wang, and Shahriar Nirjon. "Antler: Exploiting Task Affinity for Efficient Multitask Learning on Low-Resource Systems." *The International Conference on Embedded Wireless Systems and Networks (EWSN)*, 2024. (Core Rank B, Acceptance Rate: 22.8%)

## Short Papers & Abstracts

1. Run Wang\*, Shirley Bian\*, Xiaofan Yu, Quanling Zhao, **Le Zhang**, and Tajana Rosing. "Poster: Resource-Efficient Environmental Sound Classification Using Hyperdimensional Computing." *The ACM Conference on Embedded Networked Sensor Systems (SenSys)*, 2024. (Core Rank A\*)
2. **Le Zhang**, Yubo Luo, and Shahriar Nirjon. "Demo Abstract: Capuchin: A Neural Network Model Generator for 16-bit Microcontrollers." *The ACM/IEEE Conference on Information Processing in Sensor Networks (IPSN)*, 2022. (Core Rank A\*)

## Manuscripts

1. **Le Zhang**, Kailai Cui, Hao Chen, Vlad Roznyatovskiy, Jianzhong Zhang, Anhong Guo, Kang G. Shin, and Ke Sun. "EventHOI: Event-Driven Energy-Efficient Hand-Object Interaction Logging on Smartglasses." Manuscript under review *SenSys* 2027.
2. **Le Zhang**, Hao Chen, Vlad Roznyatovskiy, Jianzhong Zhang, and Ke Sun. "EgoCITE: Context-Augmented Indexing and Time-Aware Retrieval for Long-Horizon Egocentric Memory." *arXiv preprint arXiv:2608.12627*, 2026. Manuscript under review *ICLR* 2027.

## Awards & Grants

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| Dean's List: University of North Carolina at Chapel Hill | 2019—2022 |
| Honorable Mentioned Award: ICPC Mid-Atlantic Regional    | 2019      |

## Research Experience

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| <b>GSRA</b> , Ambient Intelligence (AmI) Lab, University of Michigan | 2025—2026 |
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EventHOI: Event-Driven Energy-Efficient Hand-Object Interaction Logging on Smartglasses.

Advisor: Prof. Ke Sun

- Designed an event-driven sensing and communication system for energy-efficient hand-object interaction (HOI) logging on smartglasses, leveraging low-power event sensing and IMU-assisted egomotion cancellation to trigger RGB capture only during meaningful interactions.
- Developed lightweight on-device algorithms for HOI detection and saliency-guided video compression, reducing always-on backend power consumption by up to  $10.4\times$  and uplink bandwidth by  $2.7\times$  while maintaining downstream vision-language model performance.

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| <b>Student Researcher</b> , System Energy Efficiency Lab (SeeLab), UC San Diego | 2024 |
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Resource-Constrained Collaborative Environmental Sound Recognition over LPWANs.

Advisor: Prof. Tajana Rosing

- Proposed a novel cloud-assisted offloading for resource-efficient learning on low-power wide-area networks (LPWANs) like LoRa. Designed for adaptive collaboration for resource-constrained energy-harvesting systems and low-bit rate, unreliable wireless channels.
- Achieved 2.5-12.5% improvement in accuracy, 80x in energy savings, and 220x in latency reduction compared to the state-of-the-art methods.
- 12-page full paper published in *SenSys 2025*.

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| <b>Student Researcher</b> , the Systems Energy Efficiency Lab (SeeLab), UC San Diego | 2023—2024 |
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Energy Efficient Ensemble Learning on Energy-Harvesting Systems.

Advisor: Prof. Tajana Rosing

- Developed an energy-adaptive ensemble learning framework for efficient inference and training on STM32 microcontroller. Improved the reliability of energy-harvesting machine learning system in low-energy conditions up to 40%.
- Paper published in *ASP-DAC 2025*.

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| <b>Undergraduate Research Assistant</b> , Embedded Intelligence Lab, UNC-Chapel Hill | 2021—2022 |
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Exploiting Task Affinity for Efficient Multitask Learning on Low-Resource Systems.

Advisor: Prof. Shahriar Nirjon

- Developed and evaluated an efficient multitask learning framework and memory management algorithm to optimize the orders of multitask executions on low-resource embedded systems.
- Paper accepted by *EWSN 2024*.

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| <b>Undergraduate Research Assistant</b> , Embedded Intelligence Lab, UNC-Chapel Hill | 2021—2022 |
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Efficient Neural Network Implementations on 16-bit Microcontrollers.

Advisor: Prof. Shahriar Nirjon

- Developed a neural network model generator for 16-bit TI MSP430 series microcontrollers. Improved workflow from minutes to several seconds.
- Reduced more than 50% inference time and energy consumption using hardware accelerator compared to the state-of-the-arts.
- Demo published in *IPSN 2022*.

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| <b>Mentored Research</b> , UNC-Chapel Hill | 2021 |
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Remote Collaborative Physics Simulation for High School Physics Education.

Advisor: Prof. Prasun Dewan

- Developed a remote user interface coupling platform for physics simulations in high school physics education using RPCs and IPCs to synchronize physical animations remotely.

## Teaching & Mentoring

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**Research Mentoring**, Hyperdimensional Computing on Embedded Systems

Spring 2024

Mentee: Run Wang, undergraduate student in ECE at UCSD. Publication in *SenSys '24* poster.

**Teaching Assistant**, COMP 301: Foundations of Programming, UNC-Chapel Hill

Summer 2021

Assisted in course instruction under Prof. Prasun Dewan, helping students in introductory-level Java programming.